

A DETAILED PROOF DRAFT FOR A TOROIDAL ELTON–ODELL THEOREM

DRAFT PREPARED FOR CHECKING

ABSTRACT. We give a proof draft of the following complex version of Elton–Odell. If X is an infinite-dimensional complex normed space, then there are $\varepsilon > 0$ and unit vectors (x_n) in X such that

$$\|x_n - \theta x_m\| \geq 1 + \varepsilon \quad (n \neq m, |\theta| = 1).$$

The main new point is a phase-word amplification lemma: if all weakly-null block spreading models have two-point projective distance equal to 1, then those spreading models have all positive initial sums of norm 1. The final step is a positive-sum variant of the weak-null c_0 criterion of Freeman–Odell–Sari–Zheng. Since this is the only nonstandard import, the positive-sum replacement of their Lemmas 4.3 and 4.4, the transfinite rank induction, and the Cantor–Bendixson compactness step are written out below.

1. STATEMENT AND NOTATION

Throughout, X is a complex normed space unless otherwise stated, and

$$\mathbb{T} = \{\theta \in \mathbb{C} : |\theta| = 1\}.$$

For $x, y \in X$ define

$$d_{\mathbb{T}}(x, y) = \inf_{\theta \in \mathbb{T}} \|x - \theta y\|.$$

Theorem 1.1 (Toroidal Elton–Odell). *Let X be an infinite-dimensional complex normed space. Then there are $\varepsilon > 0$ and a sequence $(x_n) \subset S_X$ such that*

$$\|x_n - \theta x_m\| \geq 1 + \varepsilon \quad \text{for all } n \neq m \text{ and all } \theta \in \mathbb{T}.$$

We use standard facts about spreading models. A normalized weakly-null sequence has a subsequence generating a spreading model, and a spreading model generated by a weakly-null sequence is suppression 1-unconditional: for every finite scalar family (a_i) and every subset A of its support,

$$\left\| \sum_{i \in A} a_i e_i \right\| \leq \left\| \sum_i a_i e_i \right\|.$$

In particular, if (e_i) is normalized and suppression 1-unconditional, then

$$\|e_1 - \theta e_2\| \geq \|e_1\| = 1 \quad (\theta \in \mathbb{T}).$$

We write $E < F$ for finite subsets of $\mathbb{N}$ when $\max E < \min F$. For a sequence (x_i) and $E \in [\mathbb{N}]^{<\omega}$ put

$$x_E = \sum_{i \in E} x_i.$$

The notation $n \leq E$ means $n \leq \min E$.

2. INITIAL REDUCTIONS

First pass to the completion. Suppose that Theorem 1.1 holds in $\widehat{X}$. Choose $(u_n) \subset S_{\widehat{X}}$ and $\eta > 0$ such that $d_{\mathbb{T}}(u_n, u_m) \geq 1 + \eta$ for $n \neq m$. Choose $v_n \in X$ with $\|v_n - u_n\|$ so small that the normalizations $y_n = v_n/\|v_n\|$ satisfy $\|y_n - u_n\| < \eta/4$. Then, for $n \neq m$ and $\theta \in \mathbb{T}$,

$$\|y_n - \theta y_m\| \geq \|u_n - \theta u_m\| - \|y_n - u_n\| - \|y_m - u_m\| \geq 1 + \eta/2.$$

Thus it suffices to prove the theorem for complex Banach spaces.

If X contains a copy of ℓ_1 , the conclusion follows from James' non-distortion theorem. Indeed, in ℓ_1 ,

$$\|e_n - \theta e_m\|_1 = 2 \quad (n \neq m, \theta \in \mathbb{T}),$$

and almost isometric copies preserve a strict lower bound larger than 1.

If X contains a copy of c_0 , the conclusion also follows from non-distortion of c_0 . In c_0 set

$$u_n = e_n - \sum_{j < n} e_j, \quad n \geq 2.$$

Then $\|u_n\|_{\infty} = 1$. If $m > n$, the vector $u_n - \theta u_m$ has among its coordinates both $1 + \theta$ and $1 - \theta$. Hence

$$\|u_n - \theta u_m\|_{\infty} \geq \max\{|1 + \theta|, |1 - \theta|\} \geq \sqrt{2} \quad (\theta \in \mathbb{T}).$$

An almost isometric copy of c_0 gives a strict toroidal separation constant larger than 1.

Consequently, for the rest of the proof assume

$$(1) \quad \ell_1 \not\hookrightarrow X, \quad c_0 \not\hookrightarrow X.$$

By the standard Rosenthal–Bessaga–Pelczynski reduction, X contains a normalized weakly-null basic sequence. Let this sequence be (x_i) , and put $Y = [x_i]$. Then Y also contains neither ℓ_1 nor c_0 . For completeness: Riesz' lemma gives a separated sequence in S_X ; since ℓ_1 does not embed, Rosenthal's theorem gives a weak-Cauchy subsequence; successive differences of a non-norm-Cauchy weak-Cauchy sequence are seminormalized and weakly null; after normalizing, the Bessaga–Pelczynski selection principle gives a basic subsequence.

3. FAILURE FORCES EQUALITY IN EVERY WEAK-NULL BLOCK SPREADING MODEL

Assume, toward contradiction, that Theorem 1.1 fails in X .

Let (y_i) be a normalized weakly-null block basis of (x_i) , and let (e_i) be a spreading model generated by a subsequence of (y_i) . Since (y_i) is weakly null, (e_i) is suppression 1-unconditional. Hence

$$\|e_1 - \theta e_2\| \geq 1 \quad (\theta \in \mathbb{T}).$$

Set

$$\Delta(e) = \inf_{\theta \in \mathbb{T}} \|e_1 - \theta e_2\|.$$

We claim that failure of Theorem 1.1 implies

$$(2) \quad \Delta(e) = 1$$

for every spreading model of every normalized weakly-null block basis of (x_i) .

Indeed, suppose $\Delta(e) > 1$. Choose $\gamma > 0$ with $\Delta(e) > 1 + 2\gamma$, and let (y_{i_j}) be the subsequence generating (e_j) . By the definition of spreading model, for all sufficiently far out pairs $p < q$,

$$\|y_{i_p} - \theta y_{i_q}\| > 1 + \gamma$$

uniformly in $\theta \in \mathbb{T}$. The uniformity follows from compactness of $\mathbb{T}$ and

$$|\|u - \theta v\| - \|u - \theta' v\|| \leq |\theta - \theta'| \|v\|.$$

Thus a tail of the generating subsequence would be toroidally $(1 + \gamma)$ -separated, contrary to the assumed failure. This proves (2).

4. THE PHASE-WORD AMPLIFICATION LEMMA

We first isolate a block-realization fact.

Lemma 4.1 (Blocks of spreading models are realized by blocks). *Let (z_i) be a normalized weakly-null block basis of (x_i) generating a spreading model (e_i) . Fix $N \in \mathbb{N}$ and scalars $a_1, \dots, a_N$. Put*

$$A = \left\| \sum_{j=1}^N a_j e_j \right\|.$$

If $A > 0$, then

$$u_r = A^{-1} \sum_{j=1}^N a_j e_{N(r-1)+j}, \quad r = 1, 2, \dots,$$

is a spreading model generated by a normalized weakly-null block basis of (x_i) .

Proof. We give the usual diagonal proof. Let D_q be a finite $1/q$ -net in the polydisc $\{(b_1, \dots, b_q) : |b_i| \leq q\}$. Since (z_i) generates (e_i) , for every q there is $m(q)$ such that, whenever $m(q) \leq i_1 < \dots < i_{qN}$ and $(b_1, \dots, b_q) \in D_q$, the two norms

$$\left\| \sum_{r=1}^q b_r \sum_{j=1}^N a_j z_{i_{(r-1)N+j}} \right\| \quad \text{and} \quad \left\| \sum_{r=1}^q b_r \sum_{j=1}^N a_j e_{(r-1)N+j} \right\|$$

differ by less than $1/q$. Choose successive integers $p(r, j)$ so rapidly increasing that every finite pattern among the blocks

$$v_r = \sum_{j=1}^N a_j z_{p(r, j)}, \quad p(r, 1) < \dots < p(r, N) < p(r+1, 1),$$

satisfies these estimates for all q up to the relevant length. Then $\|v_r\| \rightarrow A$. Since $A > 0$, after deleting finitely many terms the sequence $w_r = v_r / \|v_r\|$ is a normalized block basis of (z_i) , hence is weakly null. The same estimates, with $\|v_r\|^{-1} \rightarrow A^{-1}$, show that (w_r) generates exactly the displayed sequence (u_r) . Passing from the finite nets to all bounded scalar families uses the triangle inequality. $\square$

Lemma 4.2 (Phase-word amplification). *Assume (2). Let (e_i) be a spreading model generated by a normalized weakly-null block basis of (x_i) . Then*

$$(3) \quad \left\| \sum_{j=1}^n e_j \right\| = 1 \quad (n \in \mathbb{N}).$$

Proof. First we prove that for every dyadic integer $N = 2^k$ there are phases $a_1, \dots, a_N \in \mathbb{T}$ such that

$$(4) \quad \left\| \sum_{j=1}^N a_j e_j \right\| = 1.$$

For $N = 1$ this is immediate. Assume (4) holds for N , and put

$$u_r = \sum_{j=1}^N a_j e_{N(r-1)+j}, \quad r = 1, 2, \dots$$

Since (e_i) is spreading, $\|u_r\| = 1$ for every r . By Lemma 4.1, (u_r) is a spreading model generated by a normalized weakly-null block basis of (x_i) . Therefore (2) applies:

$$\inf_{\theta \in \mathbb{T}} \|u_1 - \theta u_2\| = 1.$$

The infimum is attained because $\mathbb{T}$ is compact. Choose $\theta \in \mathbb{T}$ with $\|u_1 - \theta u_2\| = 1$. Expanding,

$$u_1 - \theta u_2 = \sum_{j=1}^N a_j e_j - \theta \sum_{j=1}^N a_j e_{N+j}.$$

Thus the length- $2N$ word

$$(a_1, \dots, a_N, -\theta a_1, \dots, -\theta a_N)$$

has norm 1. This proves (4) by induction.

Now fix $n \in \mathbb{N}$ and $\eta > 0$. Choose a dyadic N so large that among the phases $a_1, \dots, a_N$ from (4), at least n lie in an arc of Euclidean diameter smaller than η/n . Let them be $a_{j_1}, \dots, a_{j_n}$, with $j_1 < \dots < j_n$, and choose $\alpha \in \mathbb{T}$ such that $|a_{j_r} - \alpha| < \eta/n$. By suppression 1-unconditionality and (4),

$$\left\| \sum_{r=1}^n a_{j_r} e_{j_r} \right\| \leq \left\| \sum_{j=1}^N a_j e_j \right\| = 1.$$

By spreading,

$$\left\| \sum_{r=1}^n a_{j_r} e_r \right\| = \left\| \sum_{r=1}^n a_{j_r} e_{j_r} \right\| \leq 1.$$

Therefore

$$\begin{aligned} \left\| \sum_{r=1}^n \alpha e_r \right\| &\leq \left\| \sum_{r=1}^n a_{j_r} e_r \right\| + \left\| \sum_{r=1}^n (\alpha - a_{j_r}) e_r \right\| \\ &\leq 1 + \sum_{r=1}^n |\alpha - a_{j_r}| < 1 + \eta. \end{aligned}$$

Multiplying by $\bar{\alpha}$ and letting $\eta \downarrow 0$ yields $\left\| \sum_{r=1}^n e_r \right\| \leq 1$. The reverse inequality follows from the contractive first-coordinate projection. This proves (3). $\square$

5. A POSITIVE-SUM WEAK-NULL c_0 CRITERION

The next proposition is the point at which the proof differs from a direct quotation of Freeman–Odell–Sari–Zheng. Their Theorem 4.1 assumes a two-point condition which yields full signed c_0 behavior of weak-null block spreading models. We need only positive sums, and the proof below makes explicit that no hidden signed estimate is used.

Proposition 5.1 (Positive-sum weak-null c_0 criterion). *Let (x_i) be a normalized weakly-null basic sequence in a complex Banach space $Y = [x_i]$, and assume $\ell_1 \not\hookrightarrow Y$. Suppose that every spreading model (e_i) of every normalized weakly-null block basis of (x_i) satisfies*

$$(5) \quad \left\| \sum_{i=1}^n e_i \right\| = 1 \quad (n \in \mathbb{N}).$$

Then some subsequence of (x_i) is equivalent to the unit vector basis of c_0 .

5.1. The positive replacements for FOSZ Lemmas 4.3 and 4.4.

Lemma 5.2 (Positive convex consequence). *Let (e_i) be a normalized, spreading, suppression 1-unconditional sequence satisfying (5). Then, for every $B > 0$, every $n \in \mathbb{N}$, and all real scalars $0 \leq b_i \leq B$,*

$$\left\| \sum_{i=1}^n b_i e_i \right\| \leq B.$$

Proof. For every finite $A \subset \{1, \dots, n\}$, $\|\sum_{i \in A} e_i\| \leq 1$ by spreading and (5). If $0 \leq b_i \leq B$, then

$$\sum_{i=1}^n b_i e_i = \int_0^B \sum_{\{i: b_i > t\}} e_i dt$$

in the finite-dimensional span of $e_1, \dots, e_n$. Taking norms gives the claim. $\square$

We shall use Elton's near unconditionality in the following form. Passing to a subsequence of (x_i) , there is $C < \infty$ such that

$$(6) \quad \left\| \sum_{i \in E} \delta_i x_i \right\| \leq C \left\| \sum_{i \in F} \delta_i x_i \right\|$$

whenever $E \subset F \in [\mathbb{N}]^{<\omega}$ and $\delta_i = \pm 1$ for $i \in F$. This passage to a subsequence preserves the hypothesis of Proposition 5.1. In particular, with all signs positive,

$$(7) \quad \|x_E\| \leq C \|x_F\| \quad (E \subset F \in [\mathbb{N}]^{<\omega}).$$

Lemma 5.3 (Positive version of FOSZ Lemma 4.3). *Let $\mathcal{F} \subset [\mathbb{N}]^{<\omega}$ satisfy*

$$K = \sup\{\|x_E\| : E \in \mathcal{F}\} < \infty.$$

Then there is $M \in [\mathbb{N}]^\omega$ such that for every successive sequence $E_1 < E_2 < \dots$ with $E_i \in \mathcal{F} \cap [M]^{<\omega}$, the sequence (x_{E_i}) is weakly null.

Proof. If not, then already on some infinite set satisfying (6) there are successive $E_1 < E_2 < \dots$ in $\mathcal{F}$ such that (x_{E_j}) is not weakly null. Passing to a subsequence, there are $f \in B_{Y^*}$ and $\varepsilon > 0$ with $|f(x_{E_j})| > \varepsilon$ for all j . Since the values lie in the complex plane, pass to a sector of angle less than $\pi/3$ and rotate f ; then

$$(8) \quad \operatorname{Re} f(x_{E_j}) > \varepsilon/2 \quad (j \in \mathbb{N}).$$

By complex Rosenthal, and $\ell_1 \not\hookrightarrow Y$, pass further so that (x_{E_j}) is weak Cauchy. Put

$$z_j = x_{E_{2j-1}} - x_{E_{2j}}.$$

Then (z_j) is a weakly-null block basis of (x_i) . For every finite $G \subset \mathbb{N}$ with $|G| = n$, (8) and (6), with signs $+1$ on the odd blocks and -1 on the even blocks, give

$$(9) \quad n\varepsilon/2 \leq \left\| \sum_{j \in G} x_{E_{2j-1}} \right\| \leq C \left\| \sum_{j \in G} z_j \right\|.$$

The sequence (z_j) is bounded by $2K$, and the last estimate for $|G| = 1$ implies $z_j \neq 0$. Let $w_j = z_j / \|z_j\|$ and pass to a subsequence generating a spreading model (s_j) . By the proposition hypothesis, (s_j) satisfies (5). Lemma 5.2 gives

$$\left\| \sum_{r=1}^n b_r s_r \right\| \leq 2K \quad (0 \leq b_r \leq 2K).$$

For fixed n , spreading-model convergence is uniform on the compact cube $[0, 2K]^n$ by a finite-net argument. Hence for some $j_1 < \dots < j_n$ far enough out,

$$\left\| \sum_{r=1}^n z_{j_r} \right\| = \left\| \sum_{r=1}^n \|z_{j_r}\| w_{j_r} \right\| \leq 2K + 1.$$

Combining with (9) yields $n\varepsilon/2 \leq C(2K + 1)$, impossible for large n . $\square$

Definition 5.4. A family $\mathcal{F} \subset [\mathbb{N}]^{<\omega}$ is *thin* if no member is a proper initial segment of another. It is *large* in $M \in [\mathbb{N}]^\omega$ if every infinite $L \subset M$ has an initial segment belonging to $\mathcal{F}$. For a thin family $\mathcal{F}$ put

$$\mathcal{F}^I = \{G \in [\mathbb{N}]^{<\omega} : G \text{ is an initial segment of some } F \in \mathcal{F}\}.$$

Lemma 5.5 (Positive version of FOSZ Lemma 4.4). *Let $M_0 \in [\mathbb{N}]^\omega$, and let $\mathcal{F} \subset [M_0]^{<\omega}$ be thin and large in M_0 . Assume that (x_{E_i}) is weakly null for every successive sequence $E_1 < E_2 < \dots$ in $\mathcal{F}$, and that*

$$(10) \quad \limsup_{n \rightarrow \infty} \sup\{\|x_E\| : E \in \mathcal{F}, n \leq E\} \leq 1.$$

Let (η_k) be a decreasing positive null sequence. Then there are $N = (n_i) \in [M_0]^\omega$ and a thin large family

$$\mathcal{G} = \left\{ \bigcup_{i=1}^k E_i : k \in \mathbb{N}, n_k = \min E_1, E_1 < \dots < E_k, E_i \in \mathcal{F} \cap [N]^{<\omega} \right\}$$

in N such that

$$(11) \quad \sup\{\|x_E\| : E \in \mathcal{G}, n_k \leq E\} \leq 1 + \eta_k \quad (k \in \mathbb{N}).$$

Proof. First pass to a subsequence of M_0 , still denoted by M_0 , such that

$$(12) \quad \liminf_{n \rightarrow \infty} \inf\{\|x_E\| : E \in [M_0]^{<\omega}, n \leq E\} \geq 1.$$

Indeed, choose $f_j \in S_{Y^*}$ with $f_j(x_j) = 1$. Since (x_i) is weakly null, a diagonal subsequence may be chosen so that $|f_p(x_q)| < \delta_p 2^{-q}$ for $p < q$, with $\delta_p \downarrow 0$. If $m = \min E$, then $|f_m(x_E)| \geq 1 - \delta_m$, which gives (12).

For each k define an open set $\mathcal{A}_k \subset [M_0]^\omega$ by declaring $M \in \mathcal{A}_k$ if whenever $E_1 < \dots < E_k$ are in $\mathcal{F}$ and $E = \bigcup_{i=1}^k E_i$ is an initial segment of M , one has

$$\|x_E\| \leq 1 + \eta_k.$$

The set $\mathcal{A}_k$ is open in $[M_0]^\omega$: because $\mathcal{F}$ is thin, an infinite set has at most one first $\mathcal{F}$ -initial segment, then at most one next such block in the remaining tail, and so on, so membership is decided by a finite initial segment. By Galvin–Prikry, choose nested infinite subsets $M_1 \supset M_2 \supset \dots$ of M_0 such that for each k either $[M_k]^\omega \subset \mathcal{A}_k$ or $[M_k]^\omega \cap \mathcal{A}_k = \emptyset$.

The second alternative is impossible. Suppose it holds for a fixed k . Let $E_1 < \dots < E_k$ be arbitrary members of $\mathcal{F} \cap [M_k]^{<\omega}$. Choose $M \in [M_k]^\omega$ whose initial segment is $E = \bigcup_{i=1}^k E_i$. Since $M \notin \mathcal{A}_k$, some successive $F_1 < \dots < F_k$ in $\mathcal{F}$ with $\bigcup_i F_i$ an initial segment of M satisfies $\|\sum_i x_{F_i}\| > 1 + \eta_k$. The tuple must equal the prescribed one: both F_1 and E_1 are initial segments of M and members of the thin family $\mathcal{F}$, hence $F_1 = E_1$; repeat on the tail. Thus

$$(13) \quad \left\| \sum_{i=1}^k x_{E_i} \right\| > 1 + \eta_k$$

for every successive k -tuple from $\mathcal{F} \cap [M_k]^{<\omega}$.

Using (10) and (12), choose a successive sequence $E_1 < E_2 < \dots$ in $\mathcal{F} \cap [M_k]^{<\omega}$ such that $\|x_{E_j}\| \rightarrow 1$. By hypothesis, (x_{E_j}) is weakly null. Put $y_j = x_{E_j}/\|x_{E_j}\|$ and pass to a spreading model (s_j) . Choose $0 < \delta < \eta_k/(4k)$. For all sufficiently large j we have $|\|x_{E_j}\| - 1| < \delta$. Hence, for every far-out successive k -tuple $j_1 < \dots < j_k$, (13) gives

$$\begin{aligned} \left\| \sum_{i=1}^k y_{j_i} \right\| &\geq \left\| \sum_{i=1}^k x_{E_{j_i}} \right\| - \left\| \sum_{i=1}^k (\|x_{E_{j_i}}\| - 1)y_{j_i} \right\| \\ &> 1 + \eta_k - k\delta > 1 + \eta_k/2. \end{aligned}$$

Passing to the spreading model gives

$$\left\| \sum_{i=1}^k s_i \right\| \geq 1 + \eta_k/2,$$

contrary to (5). Therefore $[M_k]^\omega \subset \mathcal{A}_k$ for every k .

Choose a diagonal sequence $N = (n_i)$ with $(n_i)_{i \geq k} \subset M_k$ for every k , and define $\mathcal{G}$ as above. If $E \in \mathcal{G}$ is formed from j blocks and $n_j = \min E$, then E is an initial segment of some member of $[M_j]^\omega$, so $\|x_E\| \leq 1 + \eta_j$. If also $n_k \leq E$, then $k \leq j$; since (η_j) is decreasing, $1 + \eta_j \leq 1 + \eta_k$. This gives (11).

Finally, $\mathcal{G}$ is large: if $L = (n_{i_r}) \subset N$ and $i_1 = k$, choose recursively $E_1 < \dots < E_k$ in $\mathcal{F} \cap [L]^{<\omega}$ so that their successive unions are initial segments of L ; the union belongs to $\mathcal{G}$. It is thin by the same uniqueness argument: comparable members have the same minimum n_k , hence the same number k of blocks, and the blocks agree recursively. $\square$

5.2. The transfinite induction, with rank bookkeeping. All Cantor–Bendixson closures are pointwise closures in $\{0, 1\}^\mathbb{N}$. If $\mathcal{F}$ is thin and large in an infinite set M , then $\mathcal{F}^I$ is a compact well-founded tree of finite subsets of M . We write

$$\text{srk}_M(\mathcal{F}) \geq \xi$$

if for every infinite $L \subset M$ the tree $(\mathcal{F} \cap [L]^{<\omega})^I$ has Cantor–Bendixson index at least ξ . This strong version records explicitly that all subsequent restrictions to infinite subsets preserve the advertised rank.

For $n \in M$ define the one-point section

$$\mathcal{F}(n) = \{E \in [M \cap (n, \infty)]^{<\omega} : \{n\} \cup E \in \mathcal{F}\}.$$

If $\mathcal{F}$ is thin and large in M , then every section $\mathcal{F}(n)$ is thin and large in the tail $M \cap (n, \infty)$, with the convention that the section $\{\emptyset\}$ has rank 1. The following elementary tree facts are the only rank facts used below.

Lemma 5.6 (Tree-rank bookkeeping). *Let $\mathcal{F}$ be thin and large in M .*

(i) *If $\text{srk}_M(\mathcal{F}) \geq \xi$ and $\mathcal{G}$ is obtained from $\mathcal{F}$ by the block-union operation in Lemma 5.5, then*

$$\text{srk}_N(\mathcal{G}) \geq \xi \cdot \omega.$$

(ii) *If $\text{srk}_M(\mathcal{F}) \geq \omega^\alpha$, $\beta < \alpha$, and $L \subset M$ is infinite, then there are $n \in L$ and an infinite $P \subset L \cap (n, \infty)$ such that*

$$\text{srk}_P(\mathcal{F}(n)) \geq \omega^\beta.$$

(iii) Consequently, if λ is a countable limit ordinal, $\gamma_j \nearrow \lambda$, and for each j we have $\gamma_j < \alpha_j < \lambda$ and a thin large family $\mathcal{H}_j$ of strong rank at least ω^{α_j} on a successive tail, then one may choose successive points n_j and successive infinite tails so that

$$\mathcal{H} = \bigcup_{j=1}^{\infty} \{ \{n_j\} \cup E : E \in \mathcal{H}_j(n_j) \cap [\{n_i : i > j\}]^{<\omega} \}$$

is thin and large in $N = \{n_1 < n_2 < \dots\}$ and satisfies $\text{srk}_N(\mathcal{H}) \geq \omega^\lambda$.

Proof. For (i), fix an infinite $L \subset N$. For each k , the initial-segment tree of $\mathcal{G} \cap [L]^{<\omega}$ contains the tree obtained by concatenating k successive copies of $(\mathcal{F} \cap [L']^{<\omega})^I$ on successive tails L' . Since every such copy has rank at least ξ , the k -fold concatenation has rank at least ξk . Taking the supremum over k gives $\xi \cdot \omega$.

For (ii), use the standard section formula for a compact tree T of finite subsets:

$$\text{CB}(T) = \sup_n (\text{CB}(T(n)) + 1), \quad T(n) = \{E : \{n\} \cup E \in T\}.$$

The same formula holds after restricting to any infinite tail. Suppose the asserted strong section failed for some $L = \{l_1 < l_2 < \dots\}$. We recursively choose an infinite subset $Q = \{q_1 < q_2 < \dots\} \subset L$ as follows. Having chosen $q_1 < \dots < q_{r-1}$ and a tail L_r , let $q_r = \min L_r$. Since the section at q_r does not have strong rank at least ω^β , choose an infinite $L_{r+1} \subset L_r \cap (q_r, \infty)$ such that

$$\text{CB}((\mathcal{F}(q_r) \cap [L_{r+1}]^{<\omega})^I) < \omega^\beta.$$

Then, for each r , the q_r -section of $(\mathcal{F} \cap [Q]^{<\omega})^I$ has rank $< \omega^\beta$. By the section formula,

$$\text{CB}((\mathcal{F} \cap [Q]^{<\omega})^I) \leq \omega^\beta < \omega^\alpha,$$

contradicting $\text{srk}_M(\mathcal{F}) \geq \omega^\alpha$.

For (iii), apply (ii) recursively. At stage j , inside the current tail choose n_j and a further infinite tail P_j so that $\text{srk}_{P_j}(\mathcal{H}_j(n_j)) \geq \omega^{\gamma_j}$; choose all later points from P_j . The displayed family is thin because different pieces have different first points and each section is thin. It is large because if $P \subset N$ is infinite with first point n_j , then $\mathcal{H}_j(n_j)$ is large in the tail after n_j and hence gives an initial segment of $P \setminus \{n_j\}$. Finally, for every infinite $P \subset N$, the indices j with $n_j \in P$ are unbounded, and the section of $\mathcal{H}$ at such an n_j , restricted to P , has rank at least ω^{γ_j} . The section formula therefore gives rank at least $\sup_j \omega^{\gamma_j} = \omega^\lambda$ on every infinite $P \subset N$. $\square$

Lemma 5.7 (Rank-and-estimate construction). *Fix an infinite $L \subset \mathbb{N}$ and a decreasing null sequence $0 < \eta_k \leq 1$. For every nonzero countable ordinal α there are $N = (n_k) \in [L]^\omega$ and a thin family $\mathcal{F}_\alpha \subset [N]^{<\omega}$ such that*

- (i) $\mathcal{F}_\alpha$ is large in N ;
- (ii) $\text{srk}_N(\mathcal{F}_\alpha) \geq \omega^\alpha$;
- (iii) for every k ,

$$(14) \quad \sup\{\|x_E\| : E \in \mathcal{F}_\alpha, n_k \leq E\} \leq 1 + \eta_k.$$

Proof. We prove this by induction on α .

For $\alpha = 1$, apply Lemma 5.5 to the singleton family $\mathcal{S} = \{\{j\} : j \in L\}$. The resulting family is the usual Schreier-type family

$$\{E \subset N : |E| = k \text{ and } \min E = n_k \text{ for some } k\},$$

with the estimate (14). Its strong rank is ω : on every infinite subset the one-point sections have arbitrarily large finite rank.

Assume the result known for α , and prove it for $\alpha + 1$. Apply the induction hypothesis on L with a decreasing null sequence bounded by 1, and obtain M and $\mathcal{F}_\alpha$. By the tail estimate, $\sup_{E \in \mathcal{F}_\alpha} \|x_E\| < \infty$. Lemma 5.3 gives an infinite $M' \subset M$ such that all successive positive block sums over $\mathcal{F}_\alpha \cap [M']^{<\omega}$ are weakly null. Strong rank ensures that restricting to M' leaves rank at least ω^α on every further infinite subset. The upper tail-limsup hypothesis required in Lemma 5.5 follows from the induction tail estimate; the lower tail estimate used in that lemma is supplied inside its proof by passing to a further subsequence. Apply Lemma 5.5 to $\mathcal{F}_\alpha \cap [M']^{<\omega}$ with the prescribed sequence (η_k) . The output has the desired norm estimate, and Lemma 5.6(i) gives strong rank at least $\omega^\alpha \cdot \omega = \omega^{\alpha+1}$.

Let λ be a countable limit ordinal. Choose ordinals $\gamma_j \nearrow \lambda$ and then choose α_j with $\gamma_j < \alpha_j < \lambda$. We construct nested infinite sets. Start with $L_1 = L$. At step j , apply the induction hypothesis for α_j inside L_j , with a decreasing error sequence all of whose terms are at most η_j . Let M_j and $\mathcal{H}_j$ be the resulting set and family. By Lemma 5.6(ii), choose $n_j \in M_j$ and an infinite $L_{j+1} \subset M_j \cap (n_j, \infty)$ such that

$$\text{srk}_{L_{j+1}}(\mathcal{H}_j(n_j)) \geq \omega^{\gamma_j}.$$

This recursion makes $n_1 < n_2 < \dots$. Put $N = (n_j)$ and $N_{j+1} = \{n_i : i > j\} \subset L_{j+1}$. Define

$$\mathcal{F}_\lambda = \bigcup_{j=1}^{\infty} \{ \{n_j\} \cup E : E \in \mathcal{H}_j(n_j) \cap [N_{j+1}]^{<\omega} \}.$$

This family is thin: comparable members must have the same first point and then come from the same thin section. It is large in N : if $P \subset N$ is infinite and $\min P = n_j$, then the section $\mathcal{H}_j(n_j)$ is large in N_{j+1} , so an initial segment of $P \setminus \{n_j\}$ gives a member of the j th piece. The estimate (14) follows because members of the j th piece are members of $\mathcal{H}_j$ and hence have norm at most $1 + \eta_j$; if $n_k \leq E$ and E lies in the j th piece, then $k \leq j$, so $1 + \eta_j \leq 1 + \eta_k$. Finally, Lemma 5.6(iii) gives $\text{srk}_N(\mathcal{F}_\lambda) \geq \omega^\lambda$. $\square$

5.3. Cantor–Bendixson compactness and bounded positive sums.

Lemma 5.8 (Closure-to-bounded-positive-sums). *Under the hypotheses of Proposition 5.1, some subsequence (x_{n_i}) has uniformly bounded positive subset sums:*

$$\sup \{ \left\| \sum_{i \in A} x_{n_i} \right\| : A \in [\mathbb{N}]^{<\omega} \} < \infty.$$

Proof. Apply Lemma 5.7 with $L = \mathbb{N}$ and $\eta_k = 2^{-k}$ for every nonzero countable ordinal α . We obtain families $\mathcal{F}_\alpha$ with strong rank at least ω^α and $\|x_E\| \leq 2$ for $E \in \mathcal{F}_\alpha$. If $G \in \mathcal{F}_\alpha^I$, choose $E \in \mathcal{F}_\alpha$ of which G is an initial segment. Then $G \subset E$, so (7) gives $\|x_G\| \leq 2C$. Thus each $\mathcal{F}_\alpha^I$ is contained in

$$\mathcal{K} = \{ G \in [\mathbb{N}]^{<\omega} : \|x_G\| \leq 2C \}.$$

Let K be the pointwise closure of $\{1_G : G \in \mathcal{K}\}$ in the Cantor cube $\{0, 1\}^\mathbb{N}$.

If K contained no characteristic function of an infinite subset of $\mathbb{N}$, then K would be a compact countable family of finite subsets of $\mathbb{N}$. Every countable compact Hausdorff space has countable Cantor–Bendixson index, say $\xi < \omega_1$. But K contains $\mathcal{F}_\alpha^I$ for every nonzero countable α , so its index is at least ω^α for every such α , a contradiction. Hence K contains 1_N for some infinite $N = (n_i)$.

By pointwise closure, for each k there exists $G_k \in \mathcal{K}$ with $\{n_1, \dots, n_k\} \subset G_k$. Therefore

$$\left\| \sum_{i=1}^k x_{n_i} \right\| \leq C \|x_{G_k}\| \leq 2C^2.$$

If $A \subset \{n_1, \dots, n_k\}$, another use of (7) gives

$$\|x_A\| \leq 2C^3.$$

Thus all positive subset sums on (x_{n_i}) are uniformly bounded. $\square$

Proof of Proposition 5.1. By Lemma 5.8, pass to a subsequence, still denoted (x_{n_i}) , and choose $M < \infty$ such that

$$\sup\{\|\sum_{i \in A} x_{n_i}\| : A \in [\mathbb{N}]^{<\omega}\} \leq M.$$

If $0 \leq a_i \leq 1$ and only finitely many a_i are nonzero, then

$$\sum_i a_i x_{n_i} = \int_0^1 \sum_{\{i: a_i > t\}} x_{n_i} dt,$$

so $\|\sum_i a_i x_{n_i}\| \leq M$. For real scalars split into positive and negative parts. For complex scalars split real and imaginary parts into their positive and negative parts. Hence

$$\left\| \sum_i a_i x_{n_i} \right\| \leq 4M \max_i |a_i|$$

for all finitely supported complex scalar families. The lower c_0 estimate follows because (x_{n_i}) is a normalized basic sequence, so its coordinate functionals are uniformly bounded. Therefore (x_{n_i}) is equivalent to the unit vector basis of c_0 . $\square$

6. COMPLETION OF THE PROOF

Assume Theorem 1.1 fails. By Section 3, every spreading model of every normalized weakly-null block basis of (x_i) satisfies (2). By Lemma 4.2, every such spreading model satisfies

$$\left\| \sum_{j=1}^n e_j \right\| = 1 \quad (n \in \mathbb{N}).$$

Therefore Proposition 5.1 applies to $Y = [x_i]$. It gives a subsequence of (x_i) equivalent to the unit vector basis of c_0 . Hence c_0 embeds into Y , and therefore into X , contradicting (1).

Thus Theorem 1.1 cannot fail. This proves the toroidal Elton–Odell theorem.

7. AUDIT NOTES

The proof above addresses the two places where a positive-sum variant could fail. First, Lemmas 5.3 and 5.5 replace the signed c_0 spreading-model conclusion only by the positive convex consequence in Lemma 5.2. Second, the transfinite construction is stated using strong rank, so restrictions to subsequences in Lemma 5.3 do not lose Cantor–Bendixson rank. The final compactness step is also explicit: if no infinite characteristic function lies in the closure of the bounded positive-sum family, the closure is a countable compact space and has countable Cantor–Bendixson index, contradicting the arbitrarily large ranks constructed above.

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